

|                                                                                                        |                                              |                                    |
|--------------------------------------------------------------------------------------------------------|----------------------------------------------|------------------------------------|
| <b>Course Code: CS1280</b>                                                                             |                                              |                                    |
| <b>Name of the Programme: BCA (Hons.)</b>                                                              |                                              |                                    |
| <b>Name of the Course: Python Programming for Data Analytics</b>                                       |                                              |                                    |
| <b>Semester: II</b>                                                                                    |                                              |                                    |
| <b>Course credit</b>                                                                                   | <b>No. of hours per week<br/>(L + T + P)</b> | <b>Total no. of Teaching hours</b> |
| 3                                                                                                      | 1+0+4                                        | 75                                 |
| <b>Course Objectives:</b>                                                                              |                                              |                                    |
| a) Introduce Python syntax, variables, and fundamental programming concepts.                           |                                              |                                    |
| b) Develop the ability to use Python data structures such as lists, tuples, dictionaries, and sets.    |                                              |                                    |
| c) Provide hands-on practice with file handling and data analysis libraries like NumPy and pandas.     |                                              |                                    |
| d) Teach data cleaning, preparation, visualization, and basic statistical analysis using Python tools. |                                              |                                    |

**Syllabus:**

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>Module - 1</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>No. of hours</b> |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>03</b>           |
| <b>Python Programming Essentials:</b><br>Introduction to Python: History, features, applications. Setting up the Python environment: Installing Python and Jupyter Notebook. Basic Python syntax: Variables, data types, and operators, Basic I/O operations, Control Structures: Conditional statements (if, elif, else), loops (for, while, nested loops), and control flow (break, continue, pass). Functions and Modules: Function definition, parameters, return value, recursive functions, and modular programming.                                                    |                     |
| <b>Module - 2</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>No. of hours</b> |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>03</b>           |
| <b>Data Structures, Class, objects and File Handling:</b><br>Data Structures: Lists, tuples, sets, and dictionaries, List and dictionary comprehensions, Arrays and 2D arrays: creation, manipulation, iterations, nested loops and applications in grid-based problems (e.g., Sudoku). Classes and objects, Instance variables and methods, File Handling: Reading and writing text files, Working with CSV files. Exception handling: Try and except, user defined exceptions.                                                                                              |                     |
| <b>Module - 3</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>No. of hours</b> |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>03</b>           |
| <b>Introduction to NumPy and pandas:</b><br>NumPy: Creating and working with NumPy arrays (1D, 2D, n-dimensional), Indexing, slicing, and reshaping arrays, Element-wise operations and broadcasting. Pandas: Series and DataFrame creation, Basic operations on DataFrames: indexing, slicing, filtering, Handling missing data: dropping and filling missing values. Descriptive Statistics and Inference: central tendencies: mean, median, and mode. Variability and dispersion: range, variance, standard deviation. Analysing data distributions (e.g., skew, kurtosis) |                     |

| Module - 4                                                                                                                                                                                                                                                                                                                                                                                                          | No. of hours |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                     | 03           |
| <b>Exploratory data analysis (EDA):</b><br>Data Cleaning and Preprocessing: Handling missing and duplicated data, Outlier Detection: Q1, Q3 using IQR. Encoding categorical variables (label encoding, one-hot encoding), Scaling and normalization (min-max scaling, z-score normalization). Advanced Operations with Pandas: concatenating, merging, and joining DataFrames, groupBy operations, and aggregation. |              |

| Module - 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | No. of hours |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 03           |
| <b>Data Visualization and Analysis with Python: From Basics to Case Studies</b><br><b>Matplotlib:</b> line plots, bar charts, histograms, scatter plots, customizing plots (titles, labels, and legends), subplots, figure layout, and visualizing IQR with box plots. <b>Seaborn:</b> Statistical plots: box plots, violin plots, pair plots, correlation heatmaps, customizing Seaborn plots (themes, color palettes). Introduction to Plotly. Case Studies: Exploratory data analysis (EDA) on real-world datasets, Examples from domains such as finance, healthcare, and sports. |              |

| Course outcomes                                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------|
| a) Apply Python programming fundamentals and data structures to build basic programs, handle files, and organize data.             |
| b) Implement data analysis using NumPy and Pandas, including descriptive statistics and data preprocessing techniques.             |
| c) Create data visualizations using Matplotlib and Seaborn to represent data insights effectively.                                 |
| d) Develop and present a complete data analysis project demonstrating skills in programming, data manipulation, and visualization. |

| Text Books |                                                                                                                                                    |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| 1          | McKinney, W. (2017). <i>Python for data analysis</i> (2nd ed.). O'Reilly Media. ISBN: 978-1491957660                                               |
| 2          | Zelle, J. M. (2010). <i>Python programming: An introduction to computer science</i> (2nd ed.). Franklin, Beedle & Associates. ISBN: 978-1590282755 |

| Reference books |                                                                                                                                                            |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1               | Sweigart, A. (2015). <i>Automate the boring stuff with Python: Practical programming for total beginners</i> . No Starch Press. ISBN: 978-1593279929       |
| 2               | VanderPlas, J. (2016). <i>Python data science handbook: Essential tools for working with data</i> . O'Reilly Media. ISBN: 978-1491912058                   |
| 3               | Muller, A. C., & Guido, S. (2016). <i>Introduction to machine learning with Python: A guide for data scientists</i> . O'Reilly Media. ISBN: 978-1449369415 |

| Reference Links |                                                                                                                                         |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| 1               | <a href="https://www.geeksforgeeks.org/data-analysis-with-python/">https://www.geeksforgeeks.org/data-analysis-with-python/</a>         |
| 2               | <a href="https://www.freecodecamp.org/learn/data-analysis-with-python">https://www.freecodecamp.org/learn/data-analysis-with-python</a> |
| 3               | <a href="https://onlinecourses.nptel.ac.in/noc21_cs45/preview">https://onlinecourses.nptel.ac.in/noc21_cs45/preview</a>                 |

| Lab Programs |                                                                                                                                                                                                                              | [60 Hrs] |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 1            | Setup and Environment:<br>A. Installation and configuration of Anaconda (install file in USB pendrive), Python 3.12 and Jupyter Notebook 7.1.2 in the local environment<br>Getting comfortable with Google Colab environment |          |
| 2            | Basic Programming and Syntax:<br>A. Simple calculator<br>B. Temperature converter (Celsius to Fahrenheit and vice versa)<br>C. Program to find the largest of three numbers                                                  |          |
| 3            | Functions and Recursion:<br>A. Factorial of a number using recursion<br>B. Word count program                                                                                                                                |          |
| 4            | Data Structures:<br>A. Program to perform various operations on lists (insert, delete, find, etc.)<br>B. Program to merge two dictionaries                                                                                   |          |
| 5            | File Handling:<br>A. Program to read a CSV file and perform basic analysis<br>Getting awareness about local and Google drive folder and file access, renaming, copying, and moving files                                     |          |
| 6            | NumPy Basics:<br>A. Creating and manipulating NumPy arrays<br>B. Program to perform matrix operations using NumPy                                                                                                            |          |
| 7            | Pandas Basics:<br>A. Creating a DataFrame from scratch and from a CSV file<br>B. Program to clean data (handling missing values, duplicates)                                                                                 |          |
| 8            | Descriptive Statistics:<br>Use mean(), median(), mode(), min(), max(), var(), std() from Pandas to measure central tendencies and dispersion of data                                                                         |          |
| 9            | Data Distribution:<br>Use describe(), quantile(), skew(), kurt(), value_counts() from Pandas for shape and distribution data                                                                                                 |          |
| 10           | Categorical Encoding:<br>A. Perform categorical encoding—OneHotEncoder, LabelEncoder from Python                                                                                                                             |          |
| 11           | Data Scaling:<br>Scale the data using Python: StandardScaler(), Normalizer(), and MinMaxScaler() from sklearn.preprocessing                                                                                                  |          |

|    |                                                                                                                                                                                                                               |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12 | <p>Outlier Detection:</p> <p>A. Visualize outliers using box plot from Matplotlib and Seaborn</p> <p>B. Identify outliers in datasets using Pandas DataFrame and Interquartile Range (IQR)</p>                                |
| 13 | <p>Advanced Pandas Operations:</p> <p>A. Advanced Pandas operations, GroupBy, merging, joining, concatenating DataFrames</p> <p>B. Time series analysis of stock prices (Optional case study)</p>                             |
| 14 | <p>Advanced Visualization with Seaborn:</p> <p>A. Using Seaborn to visualize data distributions</p> <p>B. Creating a pair plot for a DataFrame</p>                                                                            |
| 15 | <p>Correlation:</p> <p>A. Visualize correlation matrix using heatmap from Seaborn to enhance readability</p>                                                                                                                  |
| 16 | <p>Identifying Trends and Relationships:</p> <p>A. Identify the trend of any variable using line chart from Matplotlib and Seaborn</p> <p>B. Visualize the relationship between two correlated columns using scatter plot</p> |
| 17 | <p>Category and Numeric Relationships:</p> <p>Visualize the relationship between categorical and a numerical column using bar plot</p>                                                                                        |